



AERO4600 Automatic Flight Control Systems

Assignment 3 — 2024

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Abstract

This Report examines the lateral dynamics and gust response of an aircraft and builds on prior work [?], focusing on utilizing Matlab System Identification Toolbox as well as simulating aircraft response to gust inputs. The primary lateral modes analysed in this report are the side-slip (β) and bank angle (ϕ) of the aircraft, and are evaluated through the influence of aileron/rudder control inputs and gust disturbances. This report also evaluates the impact of sensor noise on the the system identification model and methods of improving the robustness of the system identification process. Additionally, the gust/turbulence was modelled using the Dryden Spectra method to aid in simulating the gust responses, in order to compute the maximum side-slip and bank angle perturbations. These finding contribute to modelling and understanding the aircraft's dynamics and control in the lateral motion.

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1. Introduction

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2. Q1 – Vertical Speed Transfer Function

To assess the feasibility of using the aircraft elevator (δ_e) to directly control the vertical speed (v_s), the transfer function $G_{\delta_e}^{v_s}$ was determined. Where $C = [0 \ -1 \ 0 \ V_{trim}]$, the Transfer Function for v_s was found to be (2.1).

$$G_{\delta_e}^{v_s}(s) = \frac{8.57(s - 16.89)(+12.79)(s + 0.04162)}{(s^2 + 0.03407s + 0.06989)(s^2 + 4.67s + 10.92)} \quad (2.1)$$

The stability of aircraft's control system can be determined by the location of its poles, shown in Figure 1. Poles can be seen on the right half of the s-plane, demonstrating instability.

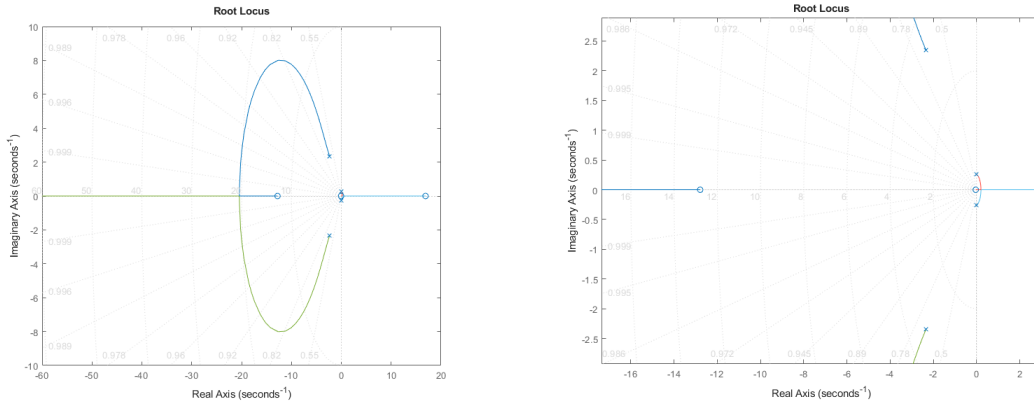


Figure 1: Root Locus of $G_{\delta_e}^{v_s}(s)$

Further, when analysing the Bode Plot of $G_{\delta_e}^{v_s}(s)$ in Figure 2, a resonant peak at a low frequency can be seen, indicating high sensitivity and possible instability. The phase response can be lagging at low frequencies. Thus, designing an autopilot using elevator to directly control vertical speed would be challenging due to the indirect relationship between elevator input and vertical speed output.

3. Q2 – Pitch Rate Autopilot

the following is for my understanding -

The primary function of the elevator is to control the pitch, which influences the angle of attack, which then in turn affects the lift produced by the wings. When the pitch angle is altered, there is a delayed response as the aircraft's trajectory adjusts to the new wing orientation. This is because the resultant lift force must change sufficiently to alter to the vertical component of the aircraft's velocity.

with no controller: settling time NaN, overshoot Inf,

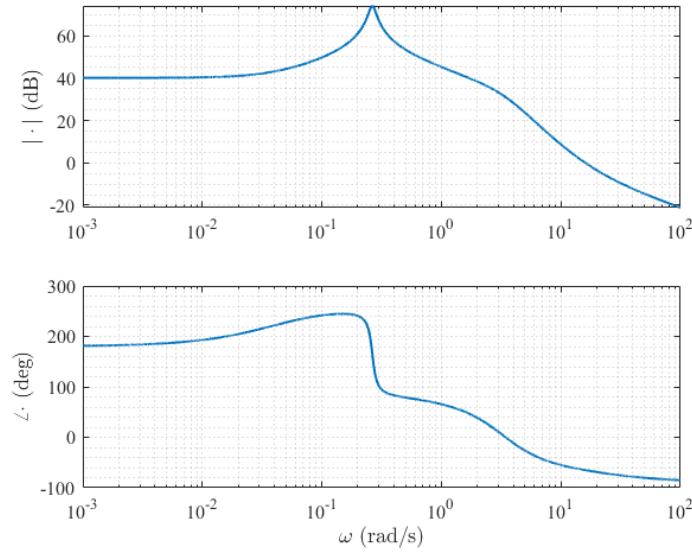


Figure 2: $G_{\delta_e}^{v_s}(s)$ Bode Plot

The Transfer function for elevator deflection to pitch rate q was found to be Equation 3.1, with root locus of that in Figure ???. The step response of the system was unstable, with 'Inf' overshoot and did not settle. Where $C = \begin{bmatrix} 0 & 0 & 1 & 0 \end{bmatrix}$,

$$G_{\delta_e}^q(s) = \frac{-26.397s(s + 1.767)(s + 0.08172)}{(s^2 + 0.03407s + 0.06989)(s^2 + 4.67s + 10.92)} \quad (3.1)$$

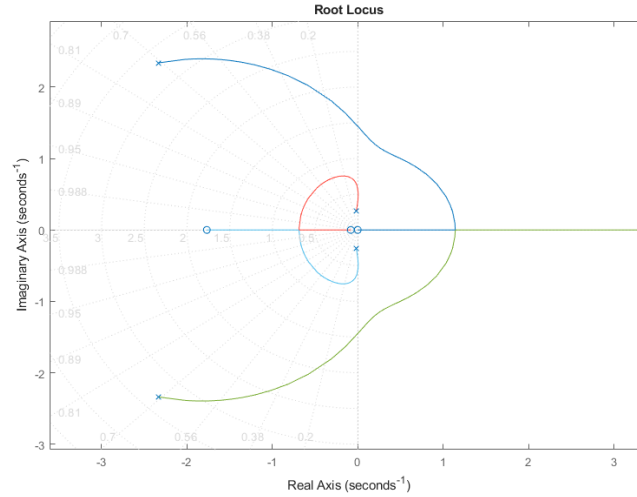


Figure 3: $G_{\delta_e}^q(s)$ Bode Plot

In order to design a controller to satisfy a bandwidth of $\omega_c = 2 - 10 \text{ rad/s}$, noise bandwidth of 20 rad/s, overshoot within 2% with settling time of around $3s$, it was determined that that system should have damping ratio (ζ) of 0.78 and Phase Margin of approximately 69.

A proportional integral derivative (PID) controller, with an additional pole, was implemented. The PID parameters were initially estimated and subsequently fine-tuned using the MATLAB 'Control System Designer' tool. Thus resulting in the controller in Equation 3.2.

$$C = \frac{-0.13015(s + 7.692)(s + 0.5263)}{s(s + 1.22e - 5)} \quad (3.2)$$

After applying the PID controller 3.2, the system had settling time of 2.7s and overshoot 8%.

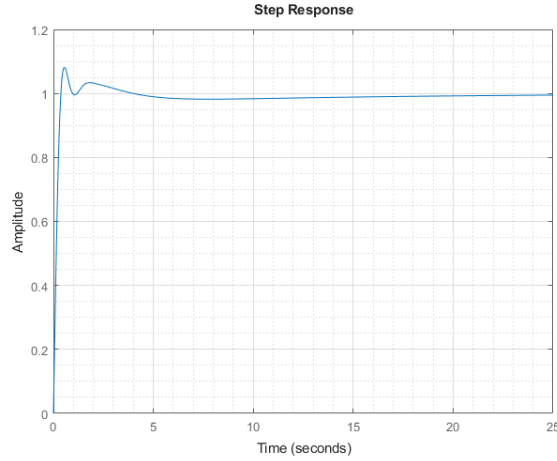


Figure 4: Step response of Controlled System

The Bode Plot in Figure 5 shows the system Phase Margin of 64.5 and frequency 5.74 rad/s, close to the desired requirements. While the designed pitch rate autopilot improved the pitch rate control performance, it did not fully meet all the specified criteria. The controller achieved desired performance in terms of settling time and Phase Margin, and effectively attenuates noise above 20 rad/s. However the response has 5% more overshoot than desired.

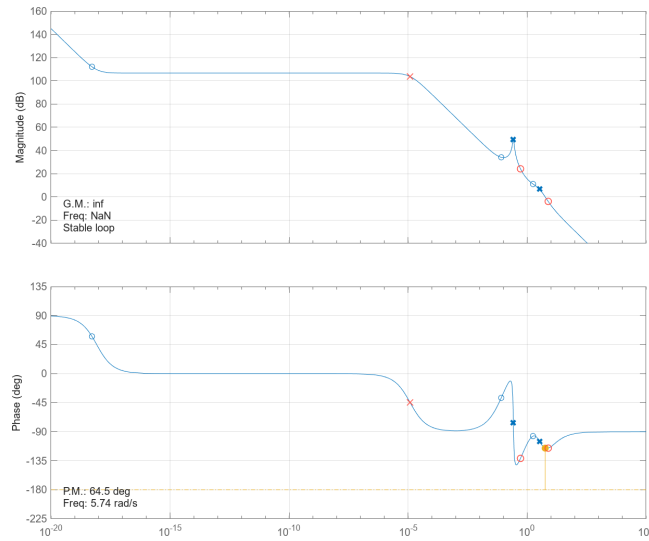


Figure 5: Bode Plot of Controlled System

4. Q3 – Vertical Speed Guidance Loop

The vertical speed guidance loop is designed to control the aircraft's ascent and descent rates by incorporating a pitch rate controller as an inner loop. This loop includes a compensator in the forward leg, which processes the vertical speed error to generate a pitch rate command. The pitch rate command then feeds into the inner loop to modulate the elevator input, allowing the system to adjust vertical speed in response to deviations from the desired reference.

4.1. Loop Topology

The complete loop topology of the vertical speed guidance loop is shown in Figure 6. The compensator transforms the vertical speed error ($vs_{ref} - vs$) into a pitch rate command. This command is then processed within the inner loop section ($de2q$), where the transfer functions G_1 and G_2 are used in the pitch rate controller to compute the required elevator deflection (δ_e) to achieve the target pitch rate.

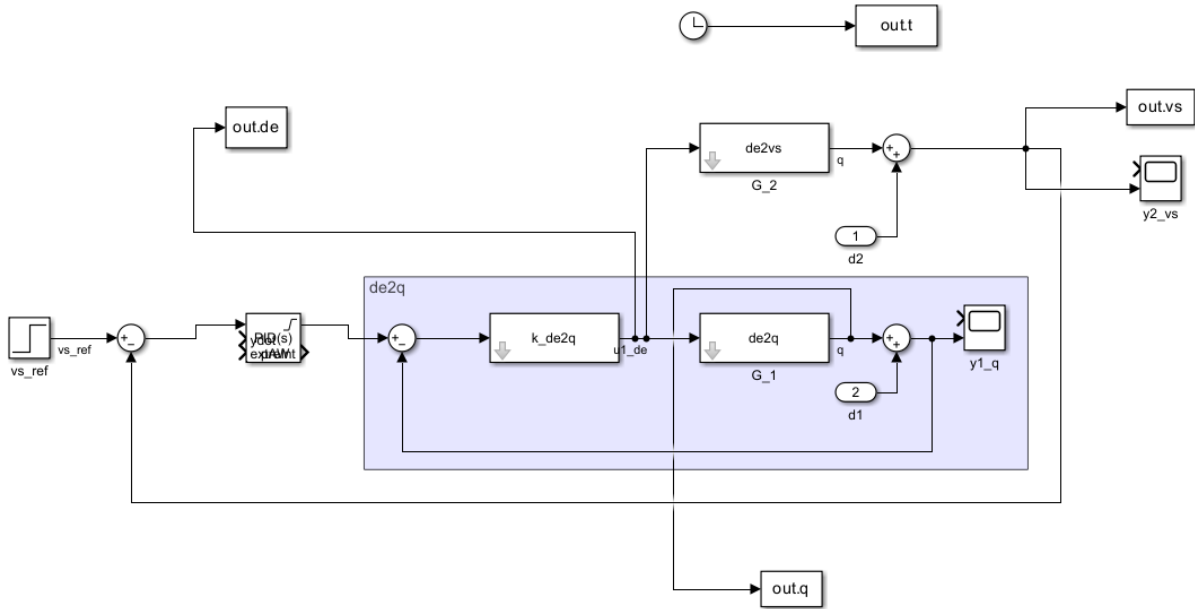


Figure 6: Vertical Speed Guidance Loop Topology

4.2. Aircraft Response to Vertical Speed Command

The vertical speed response, in Figure 7, demonstrates the ability for the system's response to a 500 ft/min (2.54 m/s) vertical speed step input command.

The response shows a rapid rise in vertical speed, reaching 9 m/s within the first 2 seconds, followed by a slight overshoot. The system then settles towards the target (8.3 m/s), achieving steady state within about 10 seconds. This response meets the desired design criteria, maintaining a rise time of about 3 seconds and settling within 10-seconds. With no oscillations and a smooth convergence towards

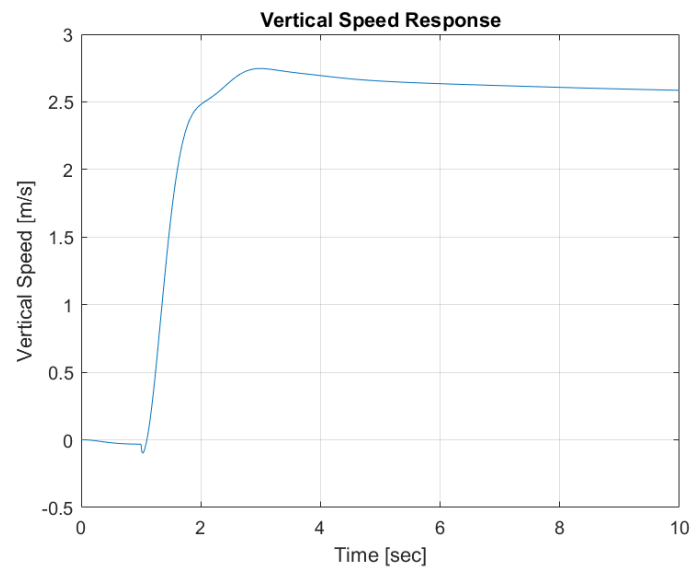


Figure 7: Vertical Speed Response

the command, and demonstrates the system's stability and damping in response to the vertical speed changes.

4.3. Pitch Rate Response

The pitch rate response, shown in Figure 8, reflects the performance of the inner pitch rate controller.

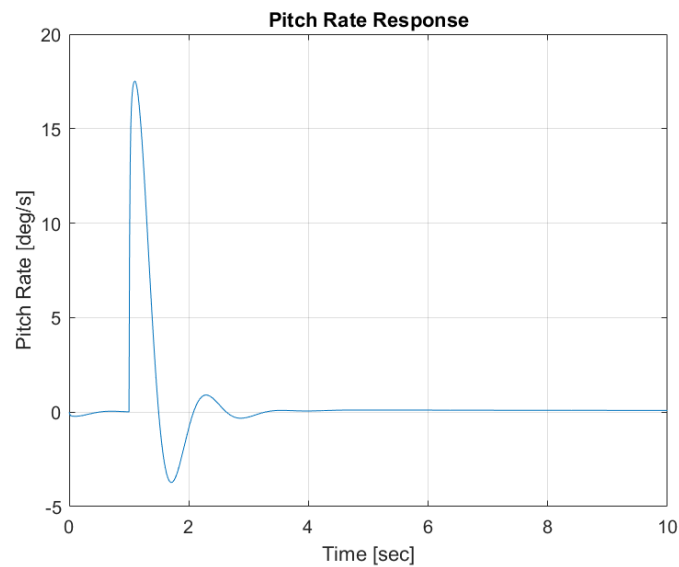


Figure 8: Pitch Rate Response

4.4. Elevator Activity

The elevator control activity is analyzed to ensure that control actions remain within acceptable operational limits.

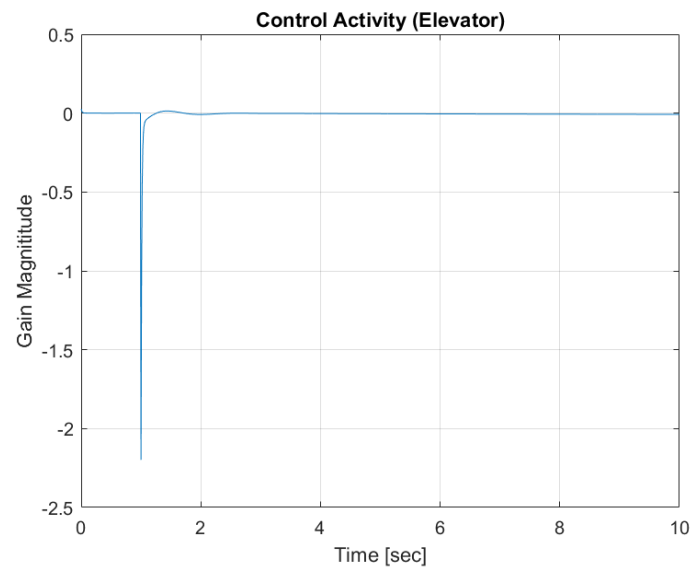


Figure 9: Elevator Control Activity

5. Q4 – Design Auto-Throttle Control System

In this section an auto-throttle design was attempted, wherein errors in airspeed would determine throttle command.

5.1. MIMO Loop Topology

The auto-throttle system was to be achieved with the MIMO loop topology shown in Figure 10.

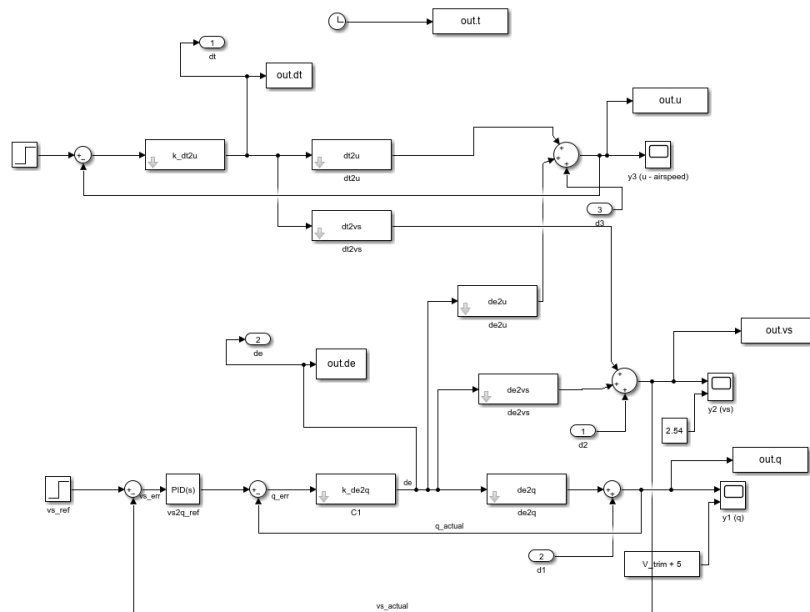


Figure 10: MIMO Simulink Loop for Auto-Throttle Design

5.2. Guidance Loop Topology Specifications

This loop was designed to the same specifications as the vertical speed guidance loop, with a vertical speed step input of 500 ft/min. The other constraints could not be implemented however due to difficulties in controlling the system. In addition to the vertical speed input there was also an input for u (airspeed).

5.3. Primary Control Effects

The primary control effects, namely the transfer function of throttle to airspeed ($(dt2u)$), and elevator to vertical speed ($de2vs$), are shown in 11 and 12.

These effects tend to follow a more sinusoidal pattern, reaching similar magnitudes over the simulation period. This is quite different from the responses found from solely the vertical speed control loop

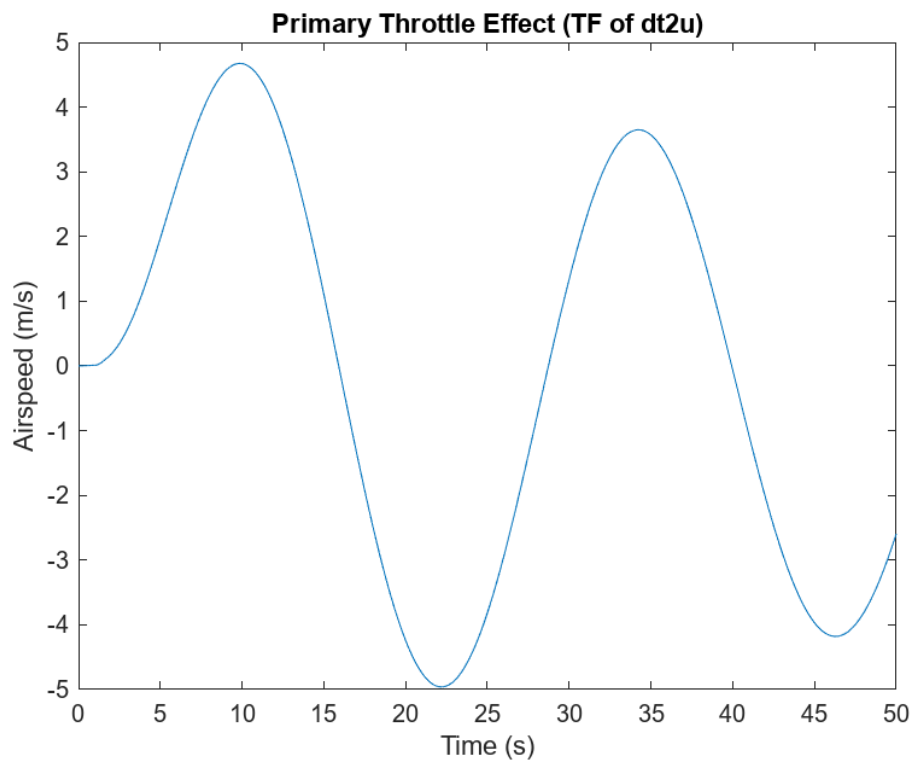


Figure 11: Primary Control Effect: Throttle to Airspeed

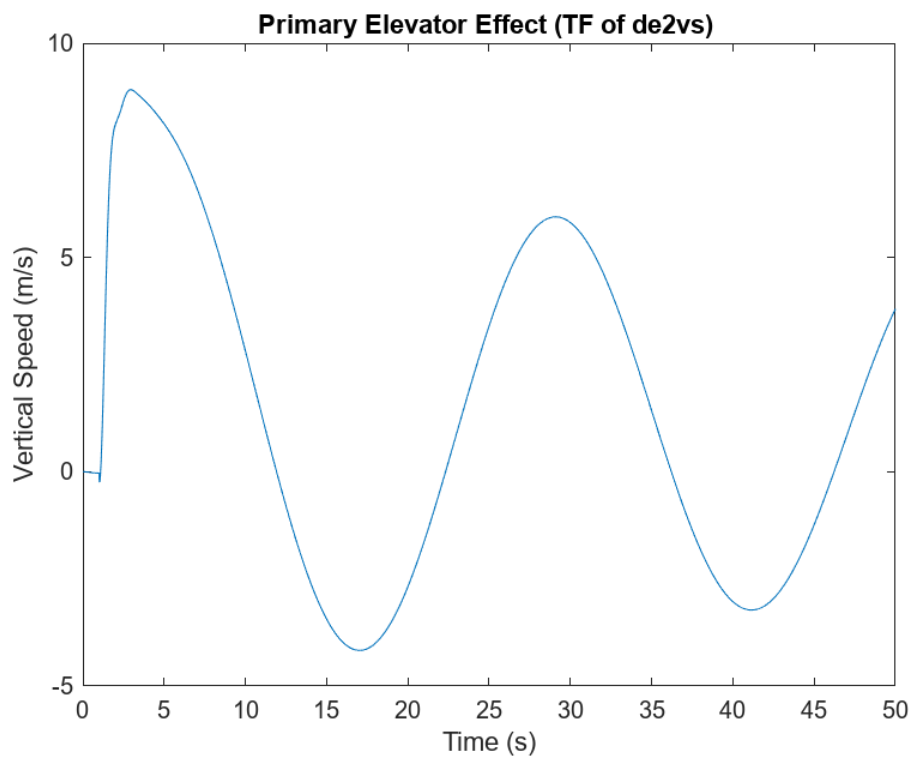


Figure 12: Primary Control Effect: Elevator to Vertical Speed

(in 4.1). This indicates the the controls have more of an effect on the outputs over time, although because of the increased complexity of the system it is more difficult to tune, and does not reach a stable point (like the outputs of the vertical speed control loop).

5.4. Cross-coupling Control Effects

The cross-coupling control effects, namely the transfer function of throttle to vertical speed ($(dt2vs)$), and elevator to airspeed ($de2u$), are shown in ?? and ??.

5.5. Elevator and Throttle Response

5.6. Auto-Throttle Effectiveness

6. Final Design

6.1. Q5 – Implementation of the control and guidance laws

6.2. Q6 – Gust Effectiveness

6.3. Q7 – Commanded Large Vertical Speed

7. Conclusion

References

A. Example of a Table